

Bus Management System



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1 Unit-Level Test Cases

1.1 User Authentication

Rules:

- Username must match an existing user in the system.
- Password must match the stored password for the username.
- User role must match the expected role for login.
- Empty username or password should be rejected.

1.1.1 Username: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
admin	Should be accepted	Has been accepted	Pass
driver1	Should be accepted	Has been accepted	Pass
nonexistent	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
user@123	Should be accepted	Has been accepted	Pass

1.1.2 Password: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
admin123	Should be accepted	Has been accepted	Pass
wrongpass	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
pass@123	Should be accepted	Has been accepted	Pass
12345	Should be accepted	Has been accepted	Pass

1.1.3 Role: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
ADMIN (for admin user)	Should be accepted	Has been accepted	Pass

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DRIVER (for driver user)	Should be accepted	Has been accepted	Pass
PASSENGER (for passenger user)	Should be accepted	Has been accepted	Pass
ADMIN (for driver user)	Should be rejected (wrong role)	Has been rejected	Pass
DRIVER (for admin user)	Should be rejected (wrong role)	Has been rejected	Pass

1.2 Bus Validation

Rules:

- Bus ID cannot be empty and must be unique
- Bus capacity must be a positive integer
- Bus model cannot be empty
- Bus status must be exactly "Active", "Maintenance", or "Inactive" (case-sensitive).

1.2.1 Bus ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
BUS001	Should be accepted	Has been accepted	Pass
BUS-001	Should be accepted	Has been accepted	Pass
123	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
B1	Should be accepted	Has been accepted	Pass

1.2.2 Bus Capacity: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
50	Should be accepted	Has been accepted	Pass
1	Should be accepted	Has been accepted	Pass
100	Should be accepted	Has been accepted	Pass
0	Should be rejected	Has been rejected	Pass
-10	Should be rejected	Has been rejected	Pass

1.2.3 Bus Model: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Volvo B7R	Should be accepted	Has been accepted	Pass
Mercedes-Benz	Should be accepted	Has been accepted	Pass
Hino	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
Bus-Model-123	Should be accepted	Has been accepted	Pass

1.2.4 Bus Status: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Active	Should be accepted	Has been accepted	Pass
Maintenance	Should be accepted	Has been accepted	Pass
Inactive	Should be accepted	Has been accepted	Pass
active	Should be rejected	Has been rejected	Pass
ACTIVE	Should be rejected	Has been rejected	Pass

1.3 Driver Validation**Rules:**

- Driver ID cannot be empty and must be unique.
- Driver name cannot be empty.

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- Contact information cannot be empty and must be at least 7 characters.
- License details cannot be empty.

1.3.1 Driver ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
DRV001	Should be accepted	Has been accepted	Pass
D-100	Should be accepted	Has been accepted	Pass
12345	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
D1	Should be accepted	Has been accepted	Pass

1.3.2 Driver Name: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
John Doe	Should be accepted	Has been accepted	Pass
Ali	Should be accepted	Has been accepted	Pass
Muhammad Hassan	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
D1	Should be accepted	Has been accepted	Pass

1.3.3 Contact Info: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
0300-1234567	Should be accepted	Has been accepted	Pass
john@email.com	Should be accepted	Has been accepted	Pass
1234567	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
123	Should be rejected	Has been rejected	Pass
" " (whitespace)	Should be rejected	Has been rejected	Pass

1.3.4 License Details: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
LIC-12345-HTV	Should be accepted	Has been accepted	Pass
Commercial-A	Should be accepted	Has been accepted	Pass
HTV-2025	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
L1	Should be accepted	Has been accepted	Pass

1.4 Route Validation:**Rules:**

- Route ID cannot be empty and must be unique.
- Origin cannot be empty.
- Destination cannot be empty.
- Origin and destination must be different.
- Estimated travel time must be a positive integer.

1.4.1 Route ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
RT001	Should be accepted	Has been accepted	Pass
Route-1	Should be accepted	Has been accepted	Pass
12345	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
R1	Should be accepted	Has been accepted	Pass

1.4.2 Origin: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Lahore	Should be accepted	Has been accepted	Pass
Karachi	Should be accepted	Has been accepted	Pass
City-A	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
Station 1	Should be accepted	Has been accepted	Pass

1.4.3 Destination: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Karachi	Should be accepted	Has been accepted	Pass
Islamabad	Should be accepted	Has been accepted	Pass
City-B	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
Station 2	Should be accepted	Has been accepted	Pass

1.4.4 Estimated Travel Time: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
60	Should be accepted	Has been accepted	Pass
1	Should be accepted	Has been accepted	Pass
480	Should be accepted	Has been accepted	Pass
0	Should be rejected	Has been rejected	Pass
-30	Should be rejected	Has been rejected	Pass

1.4.5 Origin vs Destination: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Origin: Lahore, Dest: Karachi	Should be accepted	Has been accepted	Pass
Origin: CityA, Dest: CityB	Should be accepted	Has been accepted	Pass
Origin: Lahore, Dest: Lahore	Should be rejected	Has been rejected	Pass
Origin: "", Dest: ""	Should be rejected	Has been rejected	Pass
Origin: CityA, Dest: CityA	Should be rejected	Has been rejected	Pass

1.5 Schedule Validation:**Rules:**

- Schedule ID cannot be empty and must be unique.
- Route ID, Bus ID, Driver ID must all be provided and exist in system.
- Date cannot be empty and must be in YYYY-MM-DD format.
- Departure time must be before arrival time.
- Same bus/driver cannot have overlapping schedules.

1.5.1 Schedule ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
SCH001	Should be accepted	Has been accepted	Pass
S-2026-001	Should be accepted	Has been accepted	Pass
12345	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
S1	Should be accepted	Has been accepted	Pass

1.5.2 Route ID Reference: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
RT001 (exists)	Should be accepted	Has been accepted	Pass
RT002 (exists)	Should be accepted	Has been accepted	Pass
INVALID (not exists)	Should be rejected	Has been rejected	Pass

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"" (empty)	Should be rejected	Has been rejected	Pass
RT999 (not exists)	Should be rejected	Has been rejected	Pass

1.5.3 Bus ID Reference: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
BUS001 (exists)	Should be accepted	Has been accepted	Pass
BUS002 (exists)	Should be accepted	Has been accepted	Pass
INVALID (not exists)	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
BUS999 (not exists)	Should be rejected	Has been rejected	Pass

1.5.4 Driver ID Reference: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
DRV001 (exists)	Should be accepted	Has been accepted	Pass
DRV002 (exists)	Should be accepted	Has been accepted	Pass
INVALID (not exists)	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
DRV999 (not exists)	Should be rejected	Has been rejected	Pass

1.5.5 Date: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
2026-01-15	Should be accepted	Has been accepted	Pass
2026-12-31	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
15-01-2026	Should be rejected	Has been rejected	Pass
2026-1-15	Should be rejected	Has been rejected	Pass
2026-13-01	Should be rejected	Has been rejected	Pass
2026-00-15	Should be rejected	Has been rejected	Pass
2026-01-32	Should be rejected	Has been rejected	Pass
1999-01-15	Should be rejected	Has been rejected	Pass

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abcd-ef-gh	Should be rejected	Has been rejected	Pass
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1.5.6 Departure Time: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
08:00	Should be accepted	Has been accepted	Pass
00:00	Should be accepted	Has been accepted	Pass
23:59	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
12:30	Should be accepted	Has been accepted	Pass

1.5.7 Arrival Time: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
10:00	Should be accepted	Has been accepted	Pass
23:59	Should be accepted	Has been accepted	Pass
14:30	Should be accepted	Has been accepted	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
18:00	Should be accepted	Has been accepted	Pass

1.5.8 Departure vs Arrival Time: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Departure: 08:00, Arrival: 10:00	Should be accepted	Has been accepted	Pass
Departure: 23:00, Arrival: 23:59	Should be accepted	Has been accepted	Pass
Departure: 10:00, Arrival: 08:00	Should be rejected	Has been rejected	Pass
Departure: 08:00, Arrival: 08:00	Should be rejected	Has been rejected	Pass
Departure: 12:00, Arrival: 11:00	Should be rejected	Has been rejected	Pass

1.6 Duplicate ID Check:

Rules:

- System must not allow adding an entity with an ID that already exists.
- System must allow adding an entity with a new unique ID.

1.6.1 Duplicate Bus ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
BUS001 (existing)	Should be rejected	Has been rejected	Pass
BUS999 (new)	Should be accepted	Has been accepted	Pass
BUS002 (existing)	Should be rejected	Has been rejected	Pass
NEWBUS (new)	Should be accepted	Has been accepted	Pass

1.6.2 Duplicate Driver ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
DRV001 (existing)	Should be rejected	Has been rejected	Pass
DRV999 (new)	Should be accepted	Has been accepted	Pass
DRV002 (existing)	Should be rejected	Has been rejected	Pass
NEWDRV (new)	Should be accepted	Has been accepted	Pass

1.6.3 Duplicate Route ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
RT001 (existing)	Should be rejected	Has been rejected	Pass
RT999 (new)	Should be accepted	Has been accepted	Pass
RT002 (existing)	Should be rejected	Has been rejected	Pass
NEWRT (new)	Should be accepted	Has been accepted	Pass

1.6.4 Duplicate Schedule ID: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
SCH001 (existing)	Should be rejected	Has been rejected	Pass
SCH999 (new)	Should be accepted	Has been accepted	Pass
SCH002 (existing)	Should be rejected	Has been rejected	Pass
NEWSCH (new)	Should be accepted	Has been accepted	Pass

1.7 Find Operations:**Rules:**

- If entity ID exists, return pointer to the object.
- If entity ID does not exist, return nullptr.
- Empty ID should return nullptr.

1.7.1 Find Bus: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
BUS001 (exists)	Should return Bus object	Has returned Bus object	Pass
BUS002 (exists)	Should return Bus object	Has returned Bus object	Pass
INVALID (not exists)	Should return nullptr	Has returned nullptr	Pass
"" (empty)	Should return nullptr	Has returned nullptr	Pass
BUS999 (not exists)	Should return nullptr	Has returned nullptr	Pass

1.7.2 Find Driver: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
DRV001 (exists)	Should return Driver object	Has returned Driver object	Pass
DRV002 (exists)	Should return Driver object	Has returned Driver object	Pass
INVALID (not exists)	Should return nullptr	Has returned nullptr	Pass
"" (empty)	Should return nullptr	Has returned nullptr	Pass
DRV999 (not exists)	Should return nullptr	Has returned nullptr	Pass

1.7.3 Find Route: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
RT001 (exists)	Should return Route object	Has returned Route object	Pass
RT002 (exists)	Should return Route object	Has returned Route object	Pass
INVALID (not exists)	Should return nullptr	Has returned nullptr	Pass
"" (empty)	Should return nullptr	Has returned nullptr	Pass
RT999 (not exists)	Should return nullptr	Has returned nullptr	Pass

1.7.4 Find Schedule: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
SCH001 (exists)	Should return Schedule object	Has been rejected	Pass
SCH002 (exists)	Should return Schedule object	Has been accepted	Pass
INVALID (not exists)	Should return nullptr	Has been rejected	Pass
"" (empty)	Should return nullptr	Has been accepted	Pass
SCH999 (not exists)	Should return nullptr		

1.8 Schedule Overlap Detection**Rules:**

- Same bus cannot be scheduled for overlapping times on the same date.
- Same driver cannot be assigned for overlapping times on the same date.
- Adjacent time slots do not overlap.
- Different dates never overlap.

1.8.1 Bus Time Overlap: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Same Bus, Same Date, 08:00-10:00 vs 09:00-11:00	Should be rejected (overlap)	Has been rejected	Pass
Same Bus, Same Date, 08:00-10:00 vs 11:00-13:00	Should be accepted	Has been accepted	Pass
Same Bus, Same Date, 08:00-10:00 vs 10:00-12:00	Should be accepted (adjacent)	Has been accepted	Pass
Same Bus, Different Date, 08:00-10:00 vs 08:00-10:00	Should be accepted	Has been accepted	Pass
Different Bus, Same Date, 08:00-10:00 vs 08:00-10:00	Should be accepted	Has been accepted	Pass

1.8.2 Driver Time Overlap: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Same Driver, Same Date, 08:00-10:00 vs 09:00-11:00	Should be rejected (overlap)	Has been rejected	Pass
Same Driver, Same Date, 08:00-10:00 vs 11:00-13:00	Should be accepted	Has been accepted	Pass
Same Driver, Same Date, 08:00-10:00 vs 10:00-12:00	Should be accepted (adjacent)	Has been accepted	Pass
Same Driver, Different Date, 08:00-10:00 vs 08:00-10:00	Should be accepted	Has been accepted	Pass
Different Driver, Same Date, 08:00-10:00 vs 08:00-10:00	Should be accepted	Has been accepted	Pass

1.9 Remove Operations**Rules:**

- Entity can only be removed if it exists in the system.
- Attempting to remove non-existent entity should fail.
- After removal, entity should no longer exist.

1.9.1 Remove Bus: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
BUS001 (exists)	Should be accepted	Has been accepted	Pass
INVALID (not exists)	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
BUS999 (not exists)	Should be rejected	Has been rejected	Pass
BUS001 (exists)	Should be accepted	Has been accepted	Pass

1.9.2 Remove Driver: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
DRV001 (exists)	Should be accepted	Has been accepted	Pass
INVALID (not exists)	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
DRV999 (not exists)	Should be rejected	Has been rejected	Pass

1.9.3 Remove Route: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
RT001 (exists)	Should be accepted	Has been accepted	Pass
INVALID (not exists)	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
RT999 (not exists)	Should be rejected	Has been rejected	Pass

1.9.4 Remove Schedule: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
SCH001 (exists)	Should be accepted	Has been accepted	Pass
INVALID (not exists)	Should be rejected	Has been rejected	Pass
"" (empty)	Should be rejected	Has been rejected	Pass
SCH999 (not exists)	Should be rejected	Has been rejected	Pass

1.10 User Role Conversion**Rules:**

- "Admin", "ADMIN", "admin" should convert to UserRole::ADMIN.
- "Driver", "DRIVER", "driver" should convert to UserRole::DRIVER..
- "Passenger", "PASSENGER", "passenger" should convert to UserRole::PASSENGER.
- Unknown strings default to UserRole::PASSENGER.

1.10.1 Admin Role: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Admin	Should return ADMIN	Has returned ADMIN	Pass
ADMIN	Should return ADMIN	Has returned ADMIN	Pass
admin	Should return ADMIN	Has returned ADMIN	Pass
ADmin	Should return PASSENGER (default)	Has returned PASSENGER	Pass
Admin	Should return ADMIN	Has returned ADMIN	Pass

1.10.2 Driver Role: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Driver	Should return DRIVER	Has returned DRIVER	Pass
DRIVER	Should return DRIVER	Has returned DRIVER	Pass
driver	Should return DRIVER	Has returned DRIVER	Pass
DRiver	Should return PASSENGER (default)	Has returned PASSENGER	Pass

1.10.3 Passenger Role: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Passenger	Should return PASSENGER	Has returned PASSENGER	Pass
PASSENGER	Should return PASSENGER	Has returned PASSENGER	Pass
passenger	Should return PASSENGER	Has returned PASSENGER	Pass
Unknown	Should return PASSENGER (default)	Has returned PASSENGER	Pass
"" (empty)	Should return PASSENGER (default)	Has returned PASSENGER	Pass

1.11 Route Search Operations**Rules:**

- Return all matching routes for the search criteria.
- Return empty list if no matches found.
- Search is case-sensitive (exact match).

1.11.1 Find Routes by Origin: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Lahore (exists)	Should return matching routes	Has returned ADMIN	Pass
Karachi (exists)	Should return matching routes	Has returned ADMIN	Pass
NonExistentCity	Should return empty list	Has returned ADMIN	Pass
"" (empty)	Should return empty list	Has returned PASSENGER	Pass
lahore (wrong case)	Should return empty list	Has returned ADMIN	Pass

1.11.2 Find Routes by Destination: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Karachi (exists)	Should return matching routes	Has returned matching routes	Pass
Islamabad (exists)	Should return matching routes	Has returned matching routes	Pass
NonExistentCity	Should return empty list	Has returned empty list	Pass
"" (empty)	Should return empty list	Has returned empty list	Pass
karachi (wrong case)	Should return empty list	Has returned empty list	Pass

1.11.3 Find Routes by Stop: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Multan (exists in keyStops)	Should return matching routes	Has returned matching routes	Pass
Faisalabad (exists in keyStops)	Should return matching routes	Has returned matching routes	Pass
NonExistentStop	Should return empty list	Has returned empty list	Pass
"" (empty)	Should return empty list	Has returned empty list	Pass
multan (wrong case)	Should return empty list	Has returned empty list	Pass

1.12 Schedule Search Operations**Rules:**

- Return all matching routes for the search criteria.
- Return empty list if no matches found.
- Search criteria must match exactly.

1.12.1 Get Schedules by Driver: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
DRV001 (has schedules)	Should return matching schedules	Has returned ADMIN	Pass
DRV002 (has schedules)	Should return matching schedules	Has returned ADMIN	Pass
DRV999 (no schedules)	Should return empty list	Has returned ADMIN	Pass
"" (empty)	Should return empty list	Has returned PASSENGER	Pass
INVALID	Should return empty list	Has returned ADMIN	Pass

1.12.2 Get Schedules by Date: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
2026-01-15 (has schedules)	Should return matching schedules	Has returned matching routes	Pass
2026-01-20 (has schedules)	Should return matching schedules	Has returned matching routes	Pass
2099-12-31 (no schedules)	Should return empty list	Has returned empty list	Pass
"" (empty)	Should return empty list	Has returned empty list	Pass
invalid-date	Should return empty list	Has returned empty list	Pass

1.12.3 Get Schedules by Bus: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
BUS001 (has schedules)	Should return matching schedules	Has returned matching routes	Pass
BUS002 (has schedules)	Should return matching schedules	Has returned matching routes	Pass
BUS999 (no schedules)	Should return empty list	Has returned empty list	Pass
"" (empty)	Should return empty list	Has returned empty list	Pass
INVALID	Should return empty list	Has returned empty list	Pass

1.13 Login Manager State**Rules:**

- Before login, isLoggedIn() returns false, getCurrentUser() returns nullptr.
- After successful login, isLoggedIn() returns true, getCurrentUser() returns User pointer.
- After logout, state resets to before login state.

1.13.1 isLoggedIn(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Before login	Should return false	Has returned false	Pass
After successful login	Should return true	Has returned true	Pass
After failed login	Should return false	Has returned false	Pass
After logout	Should return false	Has returned false	Pass
Before login	Should return false	Has returned false	Pass

1.13.2 getCurrentUser(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Before login	Should return nullptr	Has returned nullptr	Pass
After successful login	Should return User pointer	Has returned User pointer	Pass
After failed login	Should return nullptr	Has returned nullptr	Pass
After logout	Should return nullptr	Has returned nullptr	Pass
Before login	Should return nullptr	Has returned nullptr	Pass

1.14 Update Operations**Rules:**

- Entity can only be updated if it exists in the system.
- Updated data must pass all validation rules.
- Invalid update data should be rejected.

1.14.1 Update Bus: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Existing bus with valid data	Should be accepted	Has been accepted	Pass
Non-existing bus	Should be rejected	Has been rejected	Pass
Existing bus with capacity: 0	Should be rejected	Has been rejected	Pass
Existing bus with empty model	Should be rejected	Has been rejected	Pass
Existing bus with invalid status	Should be rejected	Has been rejected	Pass

1.14.2 Update Driver: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Existing driver with valid data	Should be accepted	Has been accepted	Pass
Non-existing driver	Should be rejected	Has been rejected	Pass
Existing driver with empty name	Should be rejected	Has been rejected	Pass
Existing driver with empty contact	Should be rejected	Has been rejected	Pass
Existing driver with empty license	Should be rejected	Has been rejected	Pass

1.14.3 Update Route: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Existing route with valid data	Should be accepted	Has been accepted	Pass
Non-existing route	Should be rejected	Has been rejected	Pass
Existing route with same origin/dest	Should be rejected	Has been rejected	Pass
Existing route with travel time: 0	Should be rejected	Has been rejected	Pass
Existing route with empty origin	Should be rejected	Has been rejected	Pass

1.14.4 Update Schedule: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Existing schedule with valid data	Should be accepted	Has been accepted	Pass
Non-existing schedule	Should be rejected	Has been rejected	Pass
Existing schedule with invalid route ref	Should be rejected	Has been rejected	Pass
Existing schedule with departure $\geq$ arrival	Should be rejected	Has been rejected	Pass
Existing schedule with empty date	Should be rejected	Has been rejected	Pass

1.15 Data Loader – CSV Parsing

Rules:

- splitCSV splits string by specified delimiter.
- Empty string returns empty vector.
- String without delimiter returns single element.
- parseKeyStops uses '|' as delimiter.

1.15.1 splitCSV(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
"a,b,c" with ','	Should return ["a", "b", "c"]	Has returned ["a", "b", "c"]	Pass
"" with ','	Should return empty vector	Has returned empty vector	Pass
"single" with ','	Should return ["single"]	Has returned ["single"]	Pass
"a b c" with ' '	Should return ["a", "b", "c"]	Has returned ["a", "b", "c"]	Pass
"a,,c" with ','	Should return ["a", "", "c"]	Has returned ["a", "", "c"]	Pass

1.15.2 parseKeyStops(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
"Stop1 Stop2 Stop3"	Should return ["Stop1", "Stop2", "Stop3"]	Has returned ["Stop1", "Stop2", "Stop3"]	Pass
""	Should return empty vector	Has returned empty vector	Pass
"SingleStop"	Should return ["SingleStop"]	Has returned ["SingleStop"]	Pass
"A B"	Should return ["A", "B"]	Has returned ["A", "B"]	Pass

1.16 Data Loader – File Operations

Rules:

- Valid file returns vector of loaded entities.
- Non-existent file returns empty vector.
- Error logged for invalid files.

1.16.1 loadBuses(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_file.txt	Should return vector of buses	Has returned vector of buses	Pass
nonexistent.txt	Should return empty vector	Has returned empty vector	Pass
empty_file.txt	Should return empty vector	Has returned empty vector	Pass

1.16.2 loadDrivers(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_file.txt	Should return vector of drivers	Has returned vector of drivers	Pass
nonexistent.txt	Should return empty vector	Has returned empty vector	Pass
empty_file.txt	Should return empty vector	Has returned empty vector	Pass

1.16.3 loadSchedules(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_file.txt	Should return vector of schedules	Has returned vector of drivers	Pass
nonexistent.txt	Should return empty vector	Has returned empty vector	Pass
empty_file.txt	Should return empty vector	Has returned empty vector	Pass

1.16.4 loadCredentials(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_file.txt	Should return vector of drivers	Has returned vector of drivers	Pass
nonexistent.txt	Should return empty vector	Has returned empty vector	Pass
empty_file.txt	Should return empty vector	Has returned empty vector	Pass

1.16.5 loadRoutes(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_file.txt	Should return vector of drivers	Has returned vector of drivers	Pass
nonexistent.txt	Should return empty vector	Has returned empty vector	Pass
empty_file.txt	Should return empty vector	Has returned empty vector	Pass

1.17 Data Loader – Invalid Line Handling**Rules:**

- Lines with incorrect field count should be skipped.
- Lines with invalid data types should be skipped.
- Empty lines should be skipped silently.
- Errors should be logged.

1.17.1 Invalid Bus Lines: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Line with 3 fields (missing 1)	Should skip and log error	Has skipped and logged error	Pass
Line with 5 fields (extra 1)	Should skip and log error	Has skipped and logged error	Pass
Line with capacity "abc"	Should skip and log error	Has skipped and logged error	Pass
Empty line	Should skip silently	Has skipped silently	Pass

1.17.2 Invalid Route Lines: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Line with 4 fields (missing 1)	Should skip and log error	Has skipped and logged error	Pass
Line with travel time "xyz"	Should skip and log error	Has skipped and logged error	Pass
Empty line	Should skip silently	Has skipped silently	Pass

1.18 Data Saver – File Operations

Rules:

- Valid path returns true.
- Invalid path returns false.
- Data saved in correct CSV format.

1.18.1 saveBuses(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_path.txt	Should return true	Has returned true	Pass
invalid/path/file.txt	Should return false	Has returned false	Pass

1.18.2 saveDrivers(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_path.txt	Should return true	Has returned true	Pass
invalid/path/file.txt	Should return false	Has returned false	Pass

1.18.3 saveRoutes(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_path.txt	Should return true	Has returned true	Pass
invalid/path/file.txt	Should return false	Has returned false	Pass

1.18.4 saveSchedules(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_path.txt	Should return true	Has returned true	Pass
invalid/path/file.txt	Should return false	Has returned false	Pass

1.18.5 saveCredentials(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
valid_path.txt	Should return true	Has returned true	Pass
invalid/path/file.txt	Should return false	Has returned false	Pass

1.19 Time Overlap Detection**Rules:**

- Overlap = NOT (arrival1 <= departure2 OR departure1 >= arrival2)
- Different dates never overlap.
- Adjacent slots do not overlap.
- Identical slots overlap.
- Contained slots overlap.

1.19.1 hasTimeOverlap(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
08:00-10:00 vs 09:00-11:00 (same date)	Should return true	Has returned true	Pass
08:00-10:00 vs 10:00-12:00 (same date)	Should return false	Has returned false	Pass
08:00-10:00 vs 06:00-08:00 (same date)	Should return false	Has returned false	Pass
08:00-12:00 vs 09:00-11:00 (same date)	Should return true (contained)	Has returned true	Pass
08:00-10:00 vs 08:00-10:00 (same date)	Should return true (identical)	Has returned true	Pass
08:00-10:00 vs 08:00-10:00 (different date)	Should return false	Has returned false	Pass

1.20 Getter/Setter Operations**Rules:**

- Setter stores value correctly
- Getter returns exact value that was set
- Values persist until changed

1.20.1 Bus Getters/Setters: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
setId("BUS001"), getId()	Should return "BUS001"	Has returned "BUS001"	Pass
setCapacity(50), getCapacity()	Should return 50	Has returned 50	Pass
setModel("Volvo"), getModel()	Should return "Volvo"	Has returned "Volvo"	Pass
setStatus("Active"), getStatus()	Should return "Active"	Has returned "Active"	Pass

1.20.2 Driver Getters/Setters: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
setId("DRV001"), getId()	Should return "DRV001"	Has returned "DRV001"	Pass
setName("John"), getName()	Should return "John"	Has returned "John"	Pass
setContactInfo("0300-123"), getContactInfo()	Should return "0300-123"	Has returned "0300-123"	Pass
setLicenseDetails("LIC-A"), getLicenseDetails()	Should return "LIC-A"	Has returned "LIC-A"	Pass

1.20.3 Route Getters/Setters: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
setId("RT001"), getId()	Should return "RT001"	Has returned "RT001"	Pass
setOrigin("Lahore"), getOrigin()	Should return "Lahore"	Has returned "Lahore"	Pass
setDestination("Karachi"), getDestination()	Should return "Karachi"	Has returned " Karachi "	Pass
setEstimatedTravelTime(120), getEstimatedTravelTime()	Should return 120	Has returned 120	Pass
addStop("Multan"), getKeyStops()	Should contain "Multan"	Has contained "Lahore"	Pass

1.20.4 Schedule Getters/Setters: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
setId("SCH001"), getId()	Should return "SCH001"	Has returned "SCH001"	Pass
setRouteId("RT001"), getRouteId()	Should return "RT001"	Has returned "RT001"	Pass
setBusId("BUS001"), getBusId()	Should return "BUS001"	Has returned " BUS001"	Pass
setDriverId("DRV001"), getDriverId()	Should return "DRV001"	Has returned " DRV001"	Pass
setDate("2026-01-15"), getDate()	Should return "2026-01-15"	Has returned "2026-01-15"	Pass
setDepartureTime("08:00"), getDepartureTime()	Should return "08:00"	Has returned "08:00"	Pass
setArrivalTime("10:00"), getArrivalTime()	Should return "10:00"	Has returned "10:00"	Pass

1.20.5 User Getters/Setters: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
setUsername("admin"), getUsername()	Should return "admin"	Has returned "admin"	Pass
setPassword("pass123"), getPassword()	Should return "pass123"	Has returned "pass123"	Pass
setRole(ADMIN), getRole()	Should return ADMIN	Has returned ADMIN	Pass
setRole(ADMIN), getRoleString()	Should return "Admin"	Has returned " Admin"	Pass
setRole(DRIVER), getRoleString()	Should return "Driver"	Has returned "Driver"	Pass
setRole(PASSENGER), getRoleString()	Should return "Passenger"	Has returned "Passenger"	Pass
setUsername("admin"), getUsername()	Should return "admin"	Has returned "admin"	Pass

1.21 Constructor Tests**Rules:**

- Default constructor initializes to default/empty values
- Parameterized constructor sets all values correctly
- Default User role is PASSENGER

1.21.1 Bus Constructors: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Bus() default	id="", capacity=0, model="", status=""	id="", capacity=0, model="", status=""	Pass
Bus("B1", 50, "Model", "Active")	All fields set correctly	All fields set correctly	Pass

1.21.2 Driver Constructors: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Driver() default	All fields empty	Has returned "DRV001"	Pass
Driver("D1", "Name", "Contact", "License")	All fields set correctly	Has returned "John"	Pass

1.21.3 Route Constructors: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Route() default	All fields empty, travelTime=0	All fields empty, travelTime=0	Pass
Route("R1", "A", "B", stops, 60)	All fields set correctly	All fields set correctly	Pass

1.21.4 Schedule Constructors: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Schedule() default	All fields empty	All fields empty	Pass
Schedule("S1", "R1", "B1", "D1", "2026-01-15", "08:00", "10:00")	All fields set correctly	All fields set correctly	Pass

1.21.5 User Constructors: V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
User() default	Role should be PASSENGER	Role is PASSENGER	Pass
User("admin", "pass", ADMIN)	All fields set correctly	All fields set correctly	Pass

1.22 Route Key Stops String Conversion**Rules:**

- Multiple stops joined with '|' delimiter
- Empty vector returns empty string
- Single stop returns that stop without delimiter

1.22.1 getStopsAsString(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
Stops: ["A", "B", "C"]	Should return "A B C"	Has returned "A B C"	Pass
Stops: []	Should return ""	Has returned ""	Pass
Stops: ["Single"]	Should return "Single"	Has returned "Single"	Pass
Stops: ["X", "Y"]	Should return "X Y"	Has returned "X Y"	Pass
Stops: ["A", "B", "C"]	Should return "A B C"	Has returned "A B C"	Pass

1.23 User Authentication Method**Rules:**

- Returns true only if both username AND password match
- Partial match returns false
- Empty credentials return false

1.23.1 authenticate(): V.1.0

Test Data	Expected Result	Actual Result	Pass/Fail
User("admin", "pass"), authenticate("admin", "pass")	Should return true	Has returned true	Pass
User("admin", "pass"), authenticate("admin", "wrong")	Should return false	Has returned false	Pass
User("admin", "pass"), authenticate("wrong", "pass")	Should return false	Has returned false	Pass
User("admin", "pass"), authenticate("", "")	Should return false	Has returned false	Pass
User("admin", "pass"), authenticate("wrong", "wrong")	Should return false	Has returned false	Pass

2 Integration-Level Test Cases

2.1 Admin Login and Dashboard Access

Test Case ID	INT_001	Test Case Description	Test Admin Login and Dashboard Access		
Created By	Ayan	Reviewed By	Sultan	Version	1.0

QA Tester's Log

Initial integration test for admin authentication flow

Tester's Name	Ayan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Application compiled and running
2	credentials.txt file exists with admin data
3	Admin user exists in system

S #	Test Data
1	Username = admin
2	Password = admin123
3	Role = ADMIN

Test Scenario

Verify admin can login with valid credentials and access Admin Dashboard

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Launch the application	Main menu displays with login options	As Expected	Pass
2	Select Admin Login option	Admin login screen appears	As Expected	Pass
3	Enter valid admin username	Username accepted	As Expected	Pass
4	Enter valid admin password	Password accepted	As Expected	Pass
5	Submit credentials	Login successful message displayed	As Expected	Pass
6	Verify Admin Dashboard loads	Admin Dashboard menu appears with all options	As Expected	Pass

2.2 Admin Login with Invalid Credentials

Test Case ID	INT_002	Test Case Description	Test Admin Login with Invalid Credentials		
Created By	Ayan	Reviewed By	Sultan	Version	1.0

QA Tester's Log

Test authentication rejection for invalid admin credentials

Tester's Name	Ayan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Application compiled and running
2	credentials.txt file exists

S #	Test Data
1	Username = admin
2	Password = wrongpassword
3	Role = ADMIN

Test Scenario

Verify system rejects invalid admin credentials and shows error

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Launch the application	Main menu displays	As Expected	Pass
2	Select Admin Login option	Admin login screen appears	As Expected	Pass
3	Enter valid admin username	Username accepted	As Expected	Pass
4	Enter invalid password	Password accepted	As Expected	Pass
5	Submit credentials	Error: Incorrect password displayed	As Expected	Pass
6	Verify Dashboard does not load	User remains at login screen	As Expected	Pass

2.3 Driver Login and Dashboard Access

Test Case ID	INT_003	Test Case Description	Test Driver Login and Dashboard Access		
Created By	Ayan	Reviewed By	Sultan	Version	1.0

QA Tester's Log Test driver authentication and dashboard integration

Tester's Name	Ayan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Application compiled and running
2	credentials.txt file with driver data
3	drivers.txt with driver profile
4	Driver exists in both files

S #	Test Data
1	Username = driver1
2	Password = driver123
3	Role = DRIVER
4	DriverID = DRV001

Test Scenario Verify driver can login and access Driver Dashboard with profile

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Launch the application	Main menu displays	As Expected	Pass
2	Select Driver Login option	Driver login screen appears	As Expected	Pass
3	Enter valid driver username	Username accepted	As Expected	Pass
4	Enter valid driver password	Password accepted	As Expected	Pass
5	Submit credentials	Login successful message displayed	As Expected	Pass
6	Verify Driver Dashboard loads	Driver Dashboard menu appears	As Expected	Pass
7	Select View Personal Profile	Driver profile displays correctly	As Expected	Pass

2.4 Admin Add New Bus Flow

Test Case ID	INT_004	Test Case Description	Test Admin Add New Bus Flow		
Created By	Ayan	Reviewed By	Sultan	Version	1.0

QA Tester's Log

Integration test for adding bus through Admin Dashboard

Tester's Name	Ayan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Admin Dashboard is displayed
3	buses.txt file exists
4	Bus ID does not already exist

S #	Test Data
1	Bus ID = BUS100
2	Capacity = 50
3	Model = Gujjar Tayyara
4	Status = Active

Test Scenario

Verify admin can add a new bus and data persists to file

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Buses	Bus management menu appears	As Expected	Pass
2	Select Add Bus option	Add bus form/prompts appear	As Expected	Pass
3	Enter Bus ID: BUS100	Bus ID accepted	As Expected	Pass
4	Enter Capacity: 50	Capacity accepted	As Expected	Pass
5	Enter Model: Gujjar Tayyara	Model accepted	As Expected	Pass
6	Enter Status: Active	Status accepted	As Expected	Pass
7	Confirm addition	Bus added successfully message	As Expected	Pass
8	Select View All Buses	New bus appears in list	As Expected	Pass

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9	Restart application and view buses	Bus persists after restart	As Expected	Pass
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2.5 Admin Add Duplicate Bus ID

Test Case ID	INT_005	Test Case Description	Test Admin Add Duplicate Bus ID		
Created By	Ayan	Reviewed By	Sultan	Version	1.0

QA Tester's Log

Test duplicate bus ID rejection in admin flow

Tester's Name	Ayan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	BUS001 already exists in system
3	buses.txt file exists
4	Bus ID does not already exist

S #	Test Data
1	Bus ID = BUS001 (existing)
2	Capacity = 40
3	Model = Hino
4	Status = Active

Test Scenario

Verify system rejects duplicate bus ID and shows error

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Buses	Bus management menu appears	As Expected	Pass
2	Select Add Bus option	Add bus form/prompts appear	As Expected	Pass
3	Enter existing Bus ID: BUS001	Bus ID accepted	As Expected	Pass
4	Enter Capacity: 40	Capacity accepted	As Expected	Pass
5	Enter Model: Hino	Model accepted	As Expected	Pass
6	Enter Status: Active	Status accepted	As Expected	Pass
7	Confirm addition	Error: Bus ID already exists	As Expected	Pass

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8	Select View All Buses	Original unchanged, duplicate	bus no	As Expected	Pass
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2.6 Admin Add New Driver Flow

Test Case ID	INT_006	Test Case Description	Test Admin Add New Driver Flow		
Created By	Sultan	Reviewed By	Noor	Version	1.0

QA Tester's Log

Integration test for adding driver through Admin Dashboard

Tester's Name	Sultan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Admin Dashboard is displayed
3	drivers.txt file exists
4	Driver ID does not already exist

S #	Test Data
1	Driver ID = DRV100
2	Name = Ayan Sajid
3	Contact = 0300-1234567
4	License = LIC-HTV-2025

Test Scenario

Verify admin can add a new driver and data persists

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Drivers	Driver management menu appears	As Expected	Pass
2	Select Add Driver option	Add driver form/prompts appear	As Expected	Pass
3	Enter Driver ID: DRV100	Driver ID accepted	As Expected	Pass
4	Enter Name: Ayan Sajid	Name accepted	As Expected	Pass
5	Enter Contact: 0300-1234567	Contact accepted	As Expected	Pass

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6	Enter License: LIC-HTV-2025	License accepted	As Expected	Pass
7	Confirm addition	Driver added successfully message	As Expected	Pass
8	Select View All Drivers	New driver appears in list	As Expected	Pass
9	Restart application and view drivers	Driver persists after restart	As Expected	Pass

2.7 Admin Add New Route Flow

Test Case ID	INT_007	Test Case Description	Test Admin Add New Route Flow		
Created By	Sultan	Reviewed By	Noor	Version	1.0

QA Tester's Log

Integration test for adding route through Admin Dashboard

Tester's Name	Sultan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Admin Dashboard is displayed
3	routes.txt file exists
4	Route ID does not already exist

S #	Test Data
1	Route ID = RT100
2	Origin = Lahore
3	Destination = Karachi
4	Key Stops = Multan Sukkur
5	Travel Time = 720

Test Scenario

Verify admin can add a new route with stops and data persists

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Drivers	Driver management menu appears	As Expected	Pass
2	Select Add Driver option	Add driver form/prompts appear	As Expected	Pass

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1	From Admin Dashboard, select Manage Routes	Route management menu appears	As Expected	Pass
2	Select Add Route option	Add route form/prompts appear	As Expected	Pass
3	Enter Route ID: RT100	Route ID accepted	As Expected	Pass
4	Enter Origin: Lahore	Origin accepted	As Expected	Pass
5	Enter Destination: Karachi	Destination accepted	As Expected	Pass
6	Enter Key Stops: Multan, Sukkur	Stops accepted	As Expected	Pass
7	Enter Travel Time: 720	Travel time accepted	As Expected	Pass
8	Confirm addition	Route added successfully message	As Expected	Pass
9	Select View All Routes	New route appears with stops	As Expected	Pass
10	Restart application	Route persists after restart	As Expected	Pass

2.8 Admin Add Route with Same Origin/Destination

Test Case ID	INT_008	Test Case Description	Test Admin Add Route with Same Origin/Destination		
Created By	Sultan	Reviewed By	Noor	Version	1.0

QA Tester's Log

Test route validation for same origin and destination

Tester's Name	Sultan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Admin Dashboard is displayed

S #	Test Data
1	Route ID = RT101
2	Origin = Lahore
3	Destination = Lahore
4	Travel Time = 60

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Test Scenario

Verify system rejects route where origin equals destination

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Routes	Route management menu appears	As Expected	Pass
2	Select Add Route option	Add route form/prompts appear	As Expected	Pass
3	Enter Route ID: RT101	Route ID accepted	As Expected	Pass
4	Enter Origin: Lahore	Origin accepted	As Expected	Pass
5	Enter Destination: Lahore	Destination accepted	As Expected	Pass
6	Enter Travel Time: 60	Travel time accepted	As Expected	Pass
7	Confirm addition	Error: Origin and destination must be different	As Expected	Pass
8	View All Routes	No new route added	As Expected	Pass

2.9 Admin Create Schedule with Valid References

Test Case ID	INT_009	Test Case Description	Test Admin Create Schedule with Valid References		
Created By	Sultan	Reviewed By	Noor	Version	1.0

QA Tester's Log

Integration test for schedule creation with valid bus, driver, route

Tester's Name	Sultan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Route RT001 exists
3	Bus BUS001 exists and is Active
4	Driver DRV001 exists

S #	Test Data
1	Schedule ID = SCH100
2	Route ID = RT001
3	Bus ID = BUS001
4	Driver ID = DRV001

SOFTWARE TESTING REPORT

5	No conflicting schedules

5	Date = 2026-01-15
6	Departure = 08:00
7	Arrival = 10:00

Test Scenario Verify admin can create schedule when all referenced entities exist

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Schedules	Schedule management menu appears	As Expected	Pass
2	Select Add Schedule option	Add schedule form appears	As Expected	Pass
3	Enter Schedule ID: SCH100	Schedule ID accepted	As Expected	Pass
4	Enter Route ID: RT001	Route ID validated and accepted	As Expected	Pass
5	Enter Bus ID: BUS001	Bus ID validated and accepted	As Expected	Pass
6	Enter Driver ID: DRV001	Driver ID validated and accepted	As Expected	Pass
7	Enter Date: 2026-01-15	Date accepted	As Expected	Pass
8	Enter Departure: 08:00	Departure time accepted	As Expected	Pass
9	Enter Arrival: 10:00	Arrival time accepted	As Expected	Pass
10	Confirm addition	Schedule added successfully	As Expected	Pass
11	View All Schedules	New schedule appears in list	As Expected	Pass

2.10 Admin Create Schedule with Invalid Route

Test Case ID	INT_010	Test Case Description	Test Admin Create Schedule with Invalid Route		
Created By	Sultan	Reviewed By	Noor	Version	1.0

QA Tester's Log Test schedule creation fails when route doesn't exist

SOFTWARE TESTING REPORT

Tester's Name	Sultan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Route RT999 does NOT exist
3	Bus BUS001 exists
4	Driver DRV001 exists

S #	Test Data
1	Schedule ID = SCH101
2	Route ID = RT999
3	Bus ID = BUS001
4	Driver ID = DRV001
5	Date = 2026-01-16

Test Scenario Verify schedule creation fails when referenced route doesn't exist

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Schedules	Schedule management menu appears	As Expected	Pass
2	Select Add Schedule option	Add schedule form appears	As Expected	Pass
3	Enter Schedule ID: SCH101	Schedule ID accepted	As Expected	Pass
4	Enter Route ID: RT999	Route ID accepted	As Expected	Pass
5	Enter Bus ID: BUS001	Bus ID accepted	As Expected	Pass
6	Enter Driver ID: DRV001	Driver ID accepted	As Expected	Pass
7	Enter remaining details	Details accepted	As Expected	Pass
8	Confirm addition	Error: Route ID RT999 not found	As Expected	Pass
9	View All Schedules	No new schedule added	As Expected	Pass

2.11 Admin Schedule Bus Overlap Detection

Test Case ID	INT_011	Test Case Description	Test Schedule Bus Overlap Detection		
Created By	Noor	Reviewed By	Ayan	Version	1.0

QA Tester's Log

Test that same bus cannot be scheduled for overlapping times

Tester's Name	Noor	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Existing schedule: BUS001, 2026-01-15, 08:00-10:00
3	Route RT001 exists
4	Driver DRV002 exists

S #	Test Data
1	Schedule ID = SCH102
2	Route ID = RT001
3	Bus ID = BUS001
4	Driver ID = DRV002
5	Date = 2026-01-15
6	Departure = 09:00
7	Arrival = 11:00

Test Scenario

Verify system prevents scheduling same bus for overlapping time slots

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Verify existing schedule for BUS001 on 2026-01-15, 08:00-10:00	Schedule exists	As Expected	Pass
2	From Admin Dashboard, select Manage Schedules	Schedule management menu appears	As Expected	Pass
3	Select Add Schedule option	Add schedule form appears	As Expected	Pass
4	Enter Schedule ID: SCH102	Schedule ID accepted	As Expected	Pass
5	Enter Route ID: RT001	Route ID accepted	As Expected	Pass
6	Enter Bus ID: BUS001 (same bus)	Bus ID accepted	As Expected	Pass

SOFTWARE TESTING REPORT

7	Enter Driver ID: DRV002	Driver ID accepted	As Expected	Pass
8	Enter Date: 2026-01-15 (same date)	Date accepted	As Expected	Pass
9	Enter Departure: 09:00, Arrival: 11:00 (overlapping)	Times accepted	As Expected	Pass
10	Confirm addition	Error: Bus already scheduled for overlapping time	As Expected	Pass

2.12 Schedule Driver Overlap Detection

Test Case ID	INT_012	Test Case Description	Test Schedule Bus Overlap Detection		
Created By	Noor	Reviewed By	Ayan	Version	1.0

QA Tester's Log

Test that same driver cannot be assigned for overlapping times

Tester's Name	Noor	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	Existing schedule: DRV001, 2026-01-15, 08:00-10:00
3	Route RT002 exists
4	Bus BUS002 exists

S #	Test Data
1	Schedule ID = SCH103
2	Route ID = RT002
3	Bus ID = BUS002
4	Driver ID = DRV001
5	Date = 2026-01-15
6	Departure = 09:00
7	Arrival = 11:00

Test Scenario

Verify system prevents assigning same driver for overlapping time slots

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Verify existing schedule for DRV001	Schedule exists	As Expected	Pass

SOFTWARE TESTING REPORT

	on 2026-01-15, 08:00-10:00			
2	From Admin Dashboard, select Manage Schedules	Schedule management menu appears	As Expected	Pass
3	Select Add Schedule option	Add schedule form appears	As Expected	Pass
4	Enter Schedule ID: SCH103	Schedule ID accepted	As Expected	Pass
5	Enter Route ID: RT002	Route ID accepted	As Expected	Pass
6	Enter Bus ID: BUS002 (different bus)	Bus ID accepted	As Expected	Pass
7	Enter Driver ID: DRV001 (same driver)	Driver ID accepted	As Expected	Pass
8	Enter Date: 2026-01-15 (same date)	Date accepted	As Expected	Pass
9	Enter Departure: 09:00, Arrival: 11:00 (overlapping)	Times accepted	As Expected	Pass
10	Confirm addition	Error: Driver already assigned for overlapping time	As Expected	Pass

2.13 Admin Remove Bus

Test Case ID	INT_013	Test Case Description	Test Admin Remove Bus		
Created By	Noor	Reviewed By	Ayan	Version	1.0

QA Tester's Log

Integration test for removing bus through Admin Dashboard

Tester's Name	Noor	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Admin logged in successfully
2	BUS003 exists in system

S #	Test Data
1	Bus ID = BUS003 (to delete)

Test Scenario

Verify admin can remove a bus and deletion persists

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	From Admin Dashboard, select Manage Buses	Bus management menu appears	As Expected	Pass
2	View All Buses, confirm BUS003 exists	BUS003 in list	As Expected	Pass
3	Select Remove Bus option	Remove prompt appears	As Expected	Pass
4	Enter Bus ID: BUS003	Bus ID accepted	As Expected	Pass
5	Confirm removal	Bus removed successfully	As Expected	Pass
6	View All Buses	BUS003 no longer in list	As Expected	Pass
7	Restart application	Bus remains deleted	As Expected	Pass

2.14 Role-Based Access Control

Test Case ID	INT_014	Test Case Description	Test Role-Based Access Control		
Created By	Noor	Reviewed By	Ayan	Version	1.0

QA Tester's Log Test that driver cannot access admin functions

Tester's Name	Noor	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Driver user exists
2	Admin functions require ADMIN role

S #	Test Data
1	Username = driver1
2	Password = driver123
3	Role = DRIVER

Test Scenario Verify driver cannot access admin login with driver credentials

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Launch application	Main menu displays	As Expected	Pass
2	Select Admin Login	Admin login screen appears	As Expected	Pass
3	Enter driver1 username	Username accepted	As Expected	Pass
4	Enter driver123 password	Password accepted	As Expected	Pass
5	Submit credentials	Error: Wrong role/Access denied	As Expected	Pass
6	Verify Admin Dashboard not shown	Dashboard does not load	As Expected	Pass

2.15 Application Handles Missing Data File

Test Case ID	INT_015	Test Case Description	Test Application Handles Missing Data File		
Created By	Noor	Reviewed By	Ayan	Version	1.0

QA Tester's Log

Test graceful handling when data file is missing

Tester's Name	Noor	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Remove/rename buses.txt temporarily
2	Other data files exist

S #	Test Data
1	buses.txt missing

Test Scenario

Verify application handles missing data file gracefully

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Rename buses.txt to buses.txt.bak	File unavailable	As Expected	Pass
2	Start application	Application launches	As Expected	Pass
3	Observe console output	Error message for missing file	As Expected	Pass
4	Application continues running	No crash	As Expected	Pass
5	Login as admin	Login successful	As Expected	Pass
6	View All Buses	Empty list displayed	As Expected	Pass
7	Restore buses.txt.bak to buses.txt	File restored	As Expected	Pass

3 System-Level Test Cases

3.1 Complete System Startup and Initialization

Test Case ID	SYS_001	Test Case Description	Test Complete System Startup and Initialization		
Created By	Ayan	Reviewed By	Sultan	Version	1.0

QA Tester's Log

Verify system initializes all components correctly on startup

Tester's Name	Ayan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	Application executable exists
2	All data files present in data/ folder
3	Valid data in all files

S #	Test Data
1	buses.txt with 5 buses
2	drivers.txt with 3 drivers
3	routes.txt with 4 routes
4	schedules.txt with 3 schedules

Test Scenario

Verify the complete Bus Management System starts up correctly, loads all data, and presents the main menu

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Execute the Bus Management System application	Application starts without errors	As Expected	Pass
2	Observe console output during startup	System displays loading messages	As Expected	Pass
3	Verify buses loaded message	"Loaded 5 buses" displayed	As Expected	Pass
4	Verify drivers loaded message	"Loaded 3 drivers" displayed	As Expected	Pass
5	Verify routes loaded message	"Loaded 4 routes" displayed	As Expected	Pass
6	Verify schedules loaded message	"Loaded 3 schedules" displayed	As Expected	Pass
7	Verify main menu displayed	Main menu with all options shown	As Expected	Pass

SOFTWARE TESTING REPORT

8	Verify all menu options are accessible	Admin, Driver, Passenger options visible	As Expected	Pass
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3.2 Admin Complete Route Management Workflow

Test Case ID	SYS_002	Test Case Description	Test Admin Complete Route Management Workflow		
Created By	Ayan	Reviewed By	Sultan	Version	1.0

QA Tester's Log

End-to-end test of admin managing routes

Tester's Name	Ayan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	System running
2	Admin logged in

S #	Test Data
1	Admin: admin/admin123
2	Route: RT500, Islamabad, Peshawar, Taxila Attock, 180

Test Scenario

Verify admin can perform complete route lifecycle: add, view, update, and remove a route

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Login as admin	Admin Dashboard shown	As Expected	Pass
2	Select Manage Routes	Route management menu displayed	As Expected	Pass
3	Select Add Route	Add route prompts appear	As Expected	Pass
4	Enter RT500, Islamabad, Peshawar, Taxila Attock, 180	Route added successfully	As Expected	Pass
5	Select View All Routes	RT500 appears with all details	As Expected	Pass
6	Verify key stops display	Taxila, Attock shown as stops	As Expected	Pass
7	Select Update Route	Update prompts appear	As Expected	Pass

SOFTWARE TESTING REPORT

8	Update RT500: Add new stop Nowshera	Route updated	As Expected	Pass
9	View All Routes	New stop visible	As Expected	Pass
10	Select Remove Route	Remove prompt appears	As Expected	Pass
11	Remove RT500	Route removed successfully	As Expected	Pass
12	View All Routes	RT500 no longer in list	As Expected	Pass

3.3 Passenger Complete Route Search Workflow

Test Case ID	SYS_003	Test Case Description	Test Passenger Complete Route Search Workflow		
Created By	Sultan	Reviewed By	Noor	Version	1.0

QA Tester's Log

End-to-end test of passenger using all search features

Tester's Name	Sultan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	System running
2	Multiple routes exist
3	Routes have different origins/destinations

S #	Test Data
1	Origin: Lahore
2	Destination: Karachi
3	Stop: Multan

Test Scenario

Verify passenger can view all routes and search by origin, destination, and stop without login

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Start application	Main menu displayed	As Expected	Pass
2	Select Passenger option	Passenger menu displayed	As Expected	Pass
3	Verify no login required	Menu accessible without credentials	As Expected	Pass
4	Select View All Routes	All routes displayed	As Expected	Pass

SOFTWARE TESTING REPORT

5	Verify route details shown	ID, Origin, Destination, Stops, Time visible	As Expected	Pass
6	Return to Passenger menu	Menu displayed	As Expected	Pass
7	Select Search by Origin	Origin prompt appears	As Expected	Pass
8	Enter: Lahore	Search executes	As Expected	Pass
9	Verify results	Only Lahore origin routes shown	As Expected	Pass
10	Return to Passenger menu	Menu displayed	As Expected	Pass
11	Select Search by Destination	Destination prompt appears	As Expected	Pass
12	Enter: Karachi	Search executes	As Expected	Pass
13	Verify results	Only Karachi destination routes shown	As Expected	Pass
14	Return to Passenger menu	Menu displayed	As Expected	Pass
15	Select Search by Stop	Stop prompt appears	As Expected	Pass
16	Enter: Multan	Search executes	As Expected	Pass
17	Verify results	Only routes with Multan stop shown	As Expected	Pass
18	Exit to main menu	Main menu displayed	As Expected	Pass

3.4 Passenger Complete Route Search Workflow

Test Case ID	SYS_004	Test Case Description	Test System Authentication Security		
Created By	Sultan	Reviewed By	Noor	Version	1.0

QA Tester's Log

Security test for authentication mechanisms

Tester's Name	Sultan	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	System running
2	Valid credentials exist

S #	Test Data
1	Valid Admin: admin/admin123
2	Valid Driver: driver1/driver123
3	Invalid: wronguser/wrongpass

Test Scenario

Verify system properly authenticates users and rejects unauthorized access

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Attempt Admin login with empty username	Error: Username required	As Expected	Pass
2	Attempt Admin login with empty password	Error: Password required	As Expected	Pass
3	Attempt Admin login with wrong username	Error: User not found	As Expected	Pass
4	Attempt Admin login with wrong password	Error: Incorrect password	As Expected	Pass
5	Attempt Admin login with driver credentials	Error: Access denied/Wrong role	As Expected	Pass
6	Attempt Driver login with admin credentials	Error: Access denied/Wrong role	As Expected	Pass
7	Attempt login with special characters (admin'--)	Error: User not found	As Expected	Pass
8	Login with valid admin credentials	Login successful	As Expected	Pass

SOFTWARE TESTING REPORT

9	Verify Admin Dashboard accessible	Dashboard displayed	As Expected	Pass
10	Logout	Main menu shown	As Expected	Pass
11	Login with valid driver credentials	Login successful	As Expected	Pass
12	Verify Driver Dashboard accessible	Dashboard displayed	As Expected	Pass
13	Verify admin functions not accessible	Only driver functions available	As Expected	Pass

3.5 System Performance with Large Dataset

Test Case ID	SYS_005	Test Case Description	Test System Performance with Large Dataset		
Created By	Noor	Reviewed By	Ayan	Version	1.0

QA Tester's Log Performance test with large number of entities

Tester's Name	Noor	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	System running
2	Admin logged in

S #	Test Data
1	100 buses
2	50 drivers
3	30 routes
4	200 schedules

Test Scenario Verify system performs acceptably with large dataset

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Populate data files with large dataset	Files populated	As Expected	Pass
2	Start application and measure startup time	Starts within acceptable time (<5 sec)	As Expected	Pass
3	Login as admin	Login successful	As Expected	Pass

SOFTWARE TESTING REPORT

4	View All Buses (100 buses)	List displayed within 2 seconds	As Expected	Pass
5	View All Drivers (50 drivers)	List displayed within 2 seconds	As Expected	Pass
6	View All Routes (30 routes)	List displayed within 2 seconds	As Expected	Pass
7	View All Schedules (200 schedules)	List displayed within 2 seconds	As Expected	Pass
8	Search for specific route by origin	Results displayed quickly	As Expected	Pass
9	Add new bus	Addition completes quickly	As Expected	Pass
10	Update existing bus	Update completes quickly	As Expected	Pass
11	Remove bus	Removal completes quickly	As Expected	Pass
12	Logout and exit	Application closes normally	As Expected	Pass
13	Restore normal test data	Original data restored	As Expected	Pass

3.6 System Exit and Cleanup

Test Case ID	SYS_006	Test Case Description	Test System Exit and Cleanup		
Created By	Noor	Reviewed By	Ayan	Version	1.0

QA Tester's Log

Verify system exits cleanly and saves all data

Tester's Name	Noor	Date Tested	3-Jan-2025	Test Case (Pass/Fail/Not Executed)	Pass
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S #	Prerequisites:
1	System running

S #	Test Data
1	Admin: admin/admin123
2	New Bus: BUS900

Test Scenario

Verify system saves all pending data and exits cleanly

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
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SOFTWARE TESTING REPORT

1	Start application	Application starts	As Expected	Pass
2	Login as admin	Admin Dashboard shown	As Expected	Pass
3	Add new bus BUS900	Bus added	As Expected	Pass
4	Exit application using exit option	Exit prompt/confirmation	As Expected	Pass
5	Confirm exit	Application terminates	As Expected	Pass
6	Verify buses.txt	BUS900 saved to file	As Expected	Pass
7	Restart application	Application starts	As Expected	Pass
8	Login as admin	Admin Dashboard shown	As Expected	Pass
9	View All Buses	BUS900 present	As Expected	Pass
10	Remove BUS900	Bus removed	As Expected	Pass
11	Exit application	Application terminates cleanly	As Expected	Pass
12	Verify no error logs	Clean shutdown	As Expected	Pass